

ALY 6040 Data Mining Applications

Module 4 – Technique Practice

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1. Introduction

The dataset captures various factors that could potentially influence student performance in exams. The key outcome of interest, or target variable, is the Exam Score, which ranges between 55 and 101. The dataset consists of 6607 records with 20 features representing different aspects of students' academic environment, habits, and personal circumstances.

Nature of the Business Context

This dataset is related to the educational sector, likely collected by an educational institution or research body interested in understanding the factors that contribute to student success in exams.

2. Problem Statement and Objective

The data includes both continuous and categorical features, making it suitable for supervised machine learning techniques such as classification, where the goal is to predict a student's success based on multiple attributes. The institution is seeking insights into how to improve academic performance by examining both academic and non-academic variables.

Key business goals could include:

- **Improving Exam Scores:** Identifying students at risk of underperforming by using features like low parental involvement, lack of access to resources, or limited extracurricular activities.
- **Targeted Interventions:** Based on the findings, schools or educational administrators can design interventions such as tutoring sessions, parental engagement programs, or providing access to necessary resources to improve performance.
- **Resource Allocation:** The dataset can help allocate resources effectively to students who need additional support by examining relationships between factors like Family Income, Access to Resources, and Tutoring Sessions.

The analysis of this data will provide actionable recommendations for teachers, policymakers, and administrators to enhance student outcomes and create more efficient educational systems.

3. Attributes and Features of the Dataset

This dataset contains a mix of continuous (numeric) and categorical variables. These features provide a holistic view of the factors influencing students' academic performance, specifically their exam scores.

This dataset encompasses a variety of factors that influence student exam performance, ranging from academic inputs like study hours and attendance, to external influences such as parental involvement, family income, and school type. These features provide a comprehensive overview of both personal and environmental variables that can be analyzed to improve educational strategies and student outcomes.

4. Exploratory Data Analysis

- Step 1:** Import necessary libraries and Loading the Dataset to perform the analysis
- Defining Numerical and Categorical Features:**

Numerical Features

This is done to understand the data deeper and understand details about the variables and features

- **Hours Studied:** The number of hours a student spends studying. This feature likely has a direct impact on academic performance, as students who study more may be better prepared for exams.

Mean: ~20 hours

- **Attendance:** The percentage of classes a student attended. Higher attendance may correlate with better performance due to increased exposure to lectures and learning materials.
Mean: ~80%
- **Sleep Hours:** Average hours of sleep per night. Sleep affects cognitive function and performance, so there is likely an optimal range for students.
Mean: ~7 hours
- **Previous Scores:** The score a student achieved in previous exams. This provides a measure of past academic performance, which may predict future outcomes.
Mean: ~75
- **Tutoring Sessions:** The number of additional tutoring sessions a student has attended.
Importance: More tutoring may help improve understanding of complex subjects, contributing to better exam performance.
- **Physical Activity:** The average time (in hours) a student spends on physical activities each day.
Importance: Physical activity can improve mental well-being and focus, which may enhance academic performance.
- **Exam Score:** The score the student achieved on a specific exam, used as the outcome variable for classification and prediction.
Mean: ~67

Categorical Features:

- **Parental Involvement:** The degree of involvement parents have in their child's education. Categories include "Low," "Medium," and "High." Parental involvement may positively affect student performance, with higher involvement potentially resulting in higher scores.
- **Access to Resources:** The availability of educational resources like textbooks, computers, and study materials at home. Categories include "Low," "Medium," and "High." Access to more resources is expected to improve performance.
- **Extracurricular Activities:** Indicates whether a student is involved in activities beyond academics (e.g., sports, music, clubs). Categories are "Yes" or "No." Participation in extracurricular activities may help develop social and time-management skills, but may also limit study time, affecting performance.
- **Motivation Level:** The student's motivation to succeed academically. Categories include "Low," "Medium," and "High." Highly motivated students tend to invest more effort into their studies, potentially leading to better results.
- **Internet Access:** Whether the student has internet access at home, either "Yes" or "No." Access to the internet allows students to access online resources, study materials, and tutorials, which can positively impact learning outcomes.
- **Family Income:** The student's family income level, categorized as "Low," "Medium," or "High." Students from higher-income families may have better access to resources and additional support like private tutors.
- **Teacher Quality:** Perception of the quality of teaching received, categorized as "Low," "Medium," or "High." High-quality teaching is a crucial factor in improving student performance.
- **School Type:** The type of school the student attends, categorized as "Public" or "Private." Private schools may have more resources and different teaching methods, potentially impacting performance.

- **Peer Influence:** The type of influence peers have on the student, categorized as “Negative,” “Neutral,” or “Positive.” Positive peer influence may foster healthy academic competition, while negative influence could lead to distractions.
- **Learning Disabilities:** Whether the student has a diagnosed learning disability, either "Yes" or "No." Students with learning disabilities may require additional support to reach their full academic potential.
- **Parental Education Level:** The highest level of education attained by the student's parents, categorized as “High School,” “College,” or “Postgraduate.” Parents with higher education levels may be more equipped to assist with schoolwork or guide their children academically.
- **Distance from Home:** The distance from home to school, categorized as “Near,” “Moderate,” or “Far.” Distance may affect the student’s time and energy available for study and extracurriculars.
- **Gender:** The gender of the student, categorized as “Male” or “Female.” Gender may influence how students engage with their education, although it should be interpreted cautiously to avoid stereotyping.

```
Student_df = pd.DataFrame(Student)
print(Student_df.info())
print(Student_df.describe())
```

	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	\
0	23	84	Low	High	
1	19	64	Low	Medium	
2	24	98	Medium	Medium	
3	29	89	Low	Medium	
4	19	92	Medium	Medium	
...					
	Parental_Education_Level	Distance_from_Home	Gender	Exam_Score	
0	High School	Near	Male	67	
1	College	Moderate	Female	61	
2	Postgraduate	Near	Male	74	
3	High School	Moderate	Male	71	
4	College	Near	Female	70	

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6607 entries, 0 to 6606
Data columns (total 20 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Hours_Studied                        6607 non-null   int64
1   Attendance                          6607 non-null   int64
2   Parental_Involvement                6607 non-null   object
3   Access_to_Resources                 6607 non-null   object
...
18  Gender                              6607 non-null   object
19  Exam_Score                          6607 non-null   int64
```

dtypes: int64(7), object(13)

memory usage: 1.0+ MB

None

	Hours_Studied	Attendance	Sleep_Hours	Previous_Scores	\
count	6607.000000	6607.000000	6607.000000	6607.000000	
mean	19.975329	79.977448	7.02906	75.070531	
std	5.990594	11.547475	1.46812	14.399784	
min	1.000000	60.000000	4.00000	50.000000	
25%	16.000000	70.000000	6.00000	63.000000	
50%	20.000000	80.000000	7.00000	75.000000	
75%	24.000000	90.000000	8.00000	88.000000	
max	44.000000	100.000000	10.00000	100.000000	

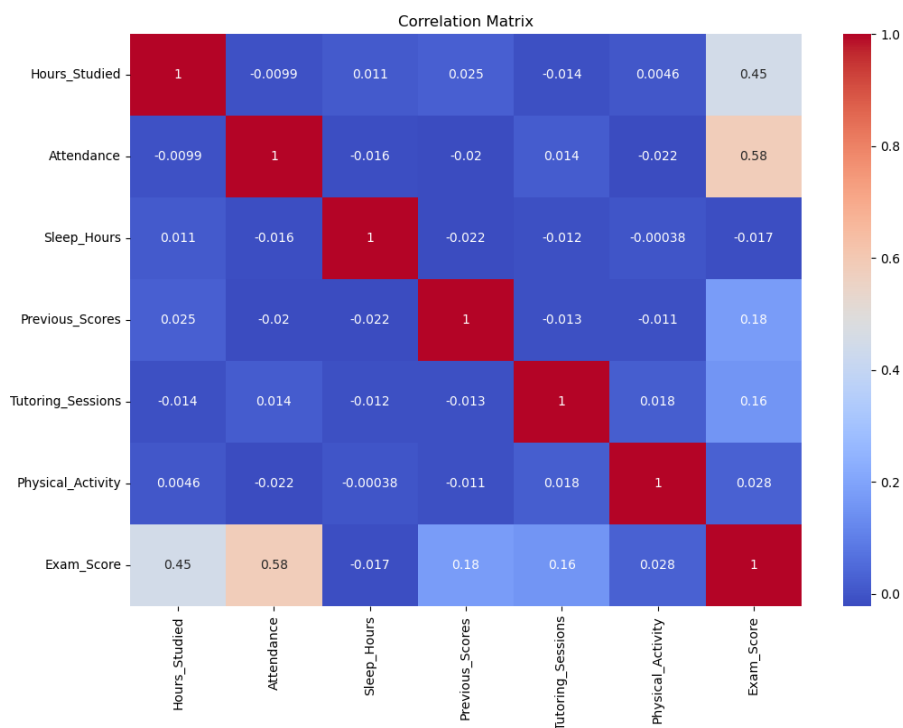
	Tutoring_Sessions	Physical_Activity	Exam_Score
count	6607.000000	6607.000000	6607.000000
mean	1.493719	2.967610	67.235659
std	1.230570	1.031231	3.890456
min	0.000000	0.000000	55.000000
25%	1.000000	2.000000	65.000000
50%	1.000000	3.000000	67.000000
75%	2.000000	4.000000	69.000000
max	8.000000	6.000000	101.000000

666678	1.000000		
max	3.000000	3.000000	2.000000

c. Correlation Matrix

The correlation matrix heatmap, is a useful way to visualize the strength and direction of relationships between numerical variables in dataset. Each cell in the matrix represents the correlation coefficient between a pair of variables, which ranges from -1 to 1.

Creating a correlation matrix is a fundamental step in exploratory data analysis (EDA). It helps identify which features are linearly related to each other and which may have significant effects on the target variable—in this case, Exam Score. Understanding the relationships between features allows to detect multicollinearity which can affect the performance of certain machine learning models like regression, identify the most important features for prediction tasks, reducing dimensionality and improving model performance and understand potential interactions between variables that can guide feature engineering or model selection.



Description :

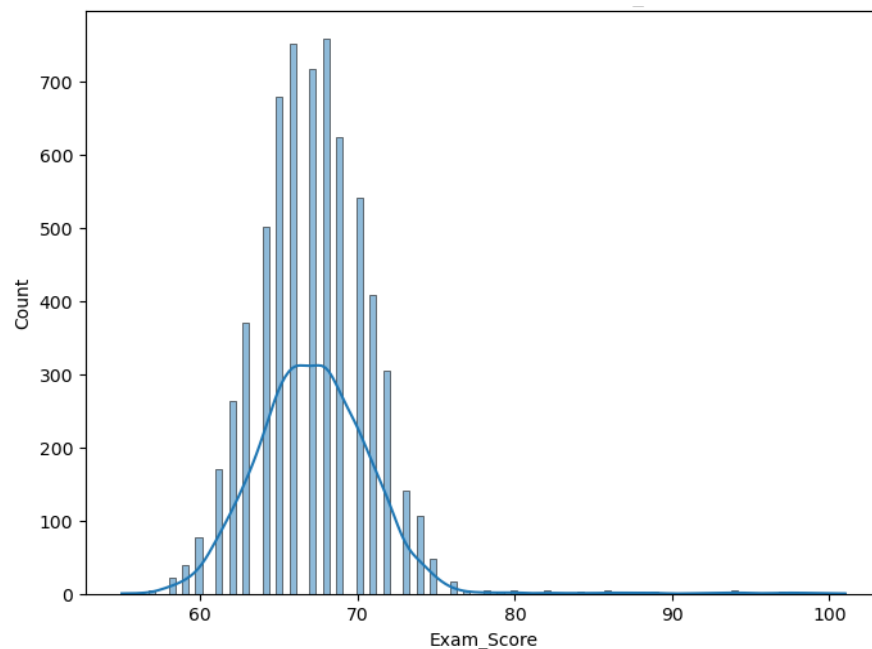
- **Hours Studied and Exam Score:** There's a moderate positive correlation (0.45), indicating that as students study more, their exam scores tend to increase.
- **Attendance and Exam Score:** A stronger positive correlation (0.58) is observed between these two variables, suggesting that students who attend classes more regularly tend to perform better on exams.
- **Previous Scores and Exam Score:** There is a mild positive correlation (0.18), implying that past performance can slightly predict future exam scores.
- **Tutoring Sessions and Exam Score:** A weak positive correlation (0.16), suggesting that attending tutoring sessions may help but isn't a dominant factor in student performance.
- **Physical Activity and Exam Score:** A very weak positive correlation (0.028), indicating that physical activity has little to no relationship with exam performance in this dataset.

- **Sleep Hours and Exam Score:** A very weak negative correlation (-0.017), showing no meaningful relationship between sleep and exam scores here.
- Attendance and study habits seem to be more influential in determining exam performance than external factors like sleep or physical activity, suggesting that efforts to improve exam scores should focus on improving these aspects.

d. Distribution of Scores

This visualization represents the distribution of Exam Score in the dataset. From the histogram and the accompanying density plot, we can observe several key details about the exam scores.

This distribution helps establish an understanding of how the target variable (exam score) is spread, which is crucial for tailoring machine learning models and making decisions about data preprocessing steps like normalization or binning of scores into categories.



Description:

- **Shape of Distribution:** The distribution is slightly left-skewed, with most students' exam scores concentrated between 60 and 75. This suggests that the majority of students performed within this range, with a gradual decline as scores increase beyond 75.
- **Peaks:** There are clear peaks between 65 and 70, where a significant portion of the data points fall. This indicates that a large number of students scored in this range, representing the "typical" exam score for this group.
- **Outliers:** There are a few outliers near the upper end of the distribution (around 100), where very few students scored close to the maximum. This small tail shows that while achieving a perfect or near-perfect score is possible, it is relatively rare.
- **Density Estimate:** The smooth line over the histogram represents a kernel density estimate (KDE), which provides a continuous view of the score distribution. This further emphasizes that most scores are clustered between 60 and 75, and the distribution has a longer tail on the right, indicating fewer high-scoring students.

- Since most scores fall between 60 and 75, focusing on the factors influencing performance within this range (e.g., study habits, attendance) may offer insights into how to improve overall student performance.

5. Data Preprocessing

a. Handle Missing Values:

- For numerical columns, missing values are replaced by the mean.
- For categorical columns, missing values are replaced by the most frequent category.

b. Encode Categorical Variables:

One-hot encoding is used to convert categorical variables into binary dummy variables. This helps clustering algorithms process categorical data.

c. Normalize or Standardize Numerical Features:

StandardScaler is applied to numerical features, which scales them to have a mean of 0 and a standard deviation of 1, ensuring all features are on the same scale.

6. Modelling and Evaluation

The Support Vector Machine (SVM) model trained here yielded an accuracy of 35.25%, which is relatively low. The confusion matrix and classification report further indicate that the model is struggling to classify several categories. Let's break down some of the key points from the classification report:

Precision, Recall, and F1-score:

Most classes have low precision, recall, and F1-scores, suggesting that the model is not performing well across different categories.

#FEATURE SCALING

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
from sklearn.svm import SVC
```

#SUPPORT VECTOR MACHINE MODEL

#Creating and Training the SVM Model

```
svm_model = SVC(kernel='linear')
svm_model.fit(X_train, y_train)
```

Predict on the test data

```
y_pred = svm_model.predict(X_test)
```

Calculate accuracy

```
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy * 100:.2f}%')
```

Confusion matrix

```
conf_matrix = confusion_matrix(y_test, y_pred)
print('Confusion Matrix:')
print(conf_matrix)
```

Classification report (includes precision, recall, F1-score)

```

class_report = classification_report(y_test, y_pred)
print('Classification Report:')
print(class_report)
Accuracy: 35.25%
Confusion Matrix:
[ 0  0  1  1  1  5  2  3  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]
[ 0  0  1  0  1  9 11  2  2  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]
[ 0  0  0  0  0 13 17 14  9  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]
[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0
  0  0  0  0  0  0  0]
...
[ 0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]
[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]
[ 0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]
[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]
[ 0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  1  0  0  0  0  0  0  0
  0  0  0  0  0  0  0]]
Classification Report:

```

	precision	recall	f1-score	support
55	0.00	0.00	0.00	1
57	0.00	0.00	0.00	2
58	0.33	0.67	0.44	3
59	0.50	0.20	0.29	5
60	0.25	0.08	0.12	13
61	0.24	0.35	0.29	26
62	0.38	0.31	0.34	54
63	0.39	0.34	0.36	77
64	0.33	0.39	0.36	94
...				
93	0.00	0.00	0.00	1
97	0.00	0.00	0.00	1
98	0.00	0.00	0.00	1
accuracy			0.35	1322
macro avg	0.22	0.23	0.21	1322
weighted avg	0.35	0.35	0.35	1322

Description:

- For some classes, precision is around 30-40%, and recall is similar, but other classes have very low performance metrics, indicating that the model is not able to generalize well for all categories.
- There may be a significant imbalance in the dataset across the target categories, which can negatively affect the SVM model's performance. SVMs typically perform better when the class distribution is more balanced.
- Feature scaling: Although we are applying feature scaling using `StandardScaler()`, it's possible that additional preprocessing might help (e.g., PCA or other dimensionality reduction techniques). The dataset could have features with complex interdependencies, making the linear kernel of the SVM insufficient to capture relationships.
- Kernel selection: The current model uses a linear kernel. Depending on the complexity of the data, a non-linear kernel (such as an RBF or polynomial kernel) might better capture the underlying patterns.
- Hyperparameter tuning: SVM models can be highly sensitive to hyperparameters like the regularization parameter `C`. Tuning these hyperparameters through methods like grid search or randomized search may improve performance.

#SUPPORT VECTOR MACHINE MODEL***#Creating and Training the SVM Model***

```
svm_model = SVC(kernel='poly')
svm_model.fit(X_train, y_train)
```

Predict on the test data

```
y_pred = svm_model.predict(X_test)
```

Calculate accuracy

```
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy * 100:.2f}%')
```

Confusion matrix

```
conf_matrix = confusion_matrix(y_test, y_pred)
print('Confusion Matrix:')
print(conf_matrix)
```

Classification report (includes precision, recall, F1-score)

```
class_report = classification_report(y_test, y_pred)
print('Classification Report:')
print(class_report)
```

Accuracy: 22.31%

accuracy			0.22	1322
macro avg	0.11	0.09	0.09	1322
weighted avg	0.22	0.22	0.21	1322

Description:

- Precision: Measures the accuracy of positive predictions. High precision indicates a low false positive rate.
- Recall: Measures the ability of the classifier to find all the positive samples. High recall indicates a low false negative rate.
- F1-score: The harmonic mean of precision and recall. It balances both metrics, especially in cases of class imbalance.
- Support: The number of actual occurrences of each class in the dataset.
- Zero Precision and Recall for Many Classes: Many classes have zero precision and recall, which suggests the model did not predict these classes at all. This is a common issue when the classes are imbalanced or if the model is not adequately complex to learn the patterns in the data.
- Overall Low Performance: The accuracy is only 22%, and the macro averages for precision, recall, and F1-score are all quite low, indicating poor performance across the classes.

#SUPPORT VECTOR MACHINE MODEL**#Creating and Training the SVM Model**

```
svm_model = SVC(kernel='rbf')
svm_model.fit(X_train, y_train)
```

Predict on the test data

```
y_pred = svm_model.predict(X_test)
```

```
print(class_report)
```

```
Accuracy: 28.21%
```

accuracy			0.28	1322
macro avg	0.13	0.12	0.11	1322
weighted avg	0.27	0.28	0.27	1322

Description:

- Zero Precision and Recall for Many Classes: As in the previous results, many classes still have zero precision and recall, indicating that the model is unable to predict these classes at all.
- Improved Performance in Some Classes: Some classes, such as 65, 66, and 68, show better precision and recall compared to the previous report, suggesting the model has learned to predict some classes more effectively.
- Overall Low Performance: Despite the small improvements, the overall macro averages for precision, recall, and F1-score remain low, indicating persistent issues with class imbalance or model fit.

7. Analyses and Interpretation

Analyzing and interpreting the current classification results can provide insights into the model's performance, the data's characteristics, and areas for further exploration.

Analysis and Interpretation of Results

- **Accuracy and Overall Performance:** The model achieved an accuracy of **28.21%**, indicating that it's currently only able to predict a little over a quarter of the instances correctly. This suggests significant room for improvement.
The **macro average precision, recall, and F1-score** are all quite low, particularly the recall, which indicates that many classes are not being predicted at all.
- **Class-Specific Performance:**
 - **Zero Precision and Recall for Many Classes:** Many classes have a precision and recall of zero, implying that the model does not predict these classes correctly at all. This is especially concerning for classes with very few instances (e.g., 55, 57, 58, 60).
 - **Classes with Better Performance:** Some classes (e.g., 65, 66, 68) have improved scores in this iteration, suggesting that the model has learned certain patterns for these classes. However, the overall prediction capability remains insufficient.
- **Class Imbalance:** The presence of numerous classes with very few instances (support counts) indicates a class imbalance issue, which can lead to models biased toward majority classes. This is evident in the results where several minority classes are entirely missed by the model.

8. Recommendations

- **Analyze Feature Importance:** Use methods like **Random Forest feature importance** to identify which features contribute most to the model's predictions. This could guide us in selecting features for further analysis or engineering.
- **Correlation Analysis:** Investigate correlations between features and the target variable. This can highlight which features have a stronger relationship with the target class labels, aiding in feature selection for future models.
- **Model Comparison:** Implement other classification algorithms (e.g., Random Forest, Gradient Boosting, KNN) and compare their performance against SVM. This can help determine which model best captures the underlying patterns in the data.
- **Hyperparameter Tuning:** Perform hyperparameter tuning using techniques like grid search or random search to find optimal parameters for different models, particularly focusing on C and gamma for SVM.
- **Additional Features:**
 - **Derived Features:** Consider creating derived features based on existing variables, such as polynomial features or interaction terms, which may help capture non-linear relationships.
 - **Statistical Features:** Include statistical summaries (e.g., mean, variance, median) of continuous features grouped by classes. This could reveal useful patterns in the data.
 - **Temporal Variables:** If applicable, investigate whether time-based variables (e.g., day of the week, time of day) influence the classification outcomes.
- **External Data Sources:** Consider integrating external datasets that may provide context or additional features. For example, if this is related to education, features like student demographics, historical performance, or environmental factors could be valuable.
- **Domain-Specific Features:** In the context of this data (e.g., educational settings), investigate if there are other relevant variables such as socioeconomic status, prior educational performance, or interventions that may impact the target variable.

9. Conclusion

- We analyzed the results of the classification models using Support Vector Machines (SVM) with polynomial and radial basis function (RBF) kernels. The models exhibited limited predictive power, with accuracy rates of approximately 22% for the polynomial kernel and 28.21% for the RBF kernel. The low macro averages for precision, recall, and F1-scores highlighted the inadequacy in predicting many classes, particularly those with fewer instances.
- The results also revealed a significant class imbalance, with numerous classes having very few instances, which negatively affected model performance. This class imbalance is an important issue that must be addressed in future analyses.
- We discussed the necessity of investigating feature importance and correlations to identify which variables contribute most significantly to the predictions. Additionally, exploring different classification algorithms and performing hyperparameter tuning will be vital next steps to optimize model performance.
- Further analyses, such as visualizing class distributions, evaluating alternative metrics, and examining new or derived features, are crucial for enhancing understanding and improving classification outcomes. Considering external data sources and implementing advanced techniques could provide new insights and strengthen the models.